

Hot Water Boilers and Expansion

A closed loop of water that gets hot and therefore gets bigger. Most boiler questions come back to that.

YOU HAVE MISSED THIS 4 TIMES

A closed hot water system:

Provides an even distribution of heat.

Four times missed. Water carries far more heat than air for its volume and circulates continuously through a sealed loop, so every emitter on the circuit runs at a similar temperature. The result is even heat rather than the surges of a warm air system.

Hydronic system	Any system that moves heat using water or steam. A steam heating system is a type of hydronic system.
Forced hot water	Determined by looking for a circulating pump near the boiler. Without a pump it moves by gravity.
Tankless coil system	Coils immersed directly in the boiler water to supply domestic hot water.
Convactor	A heat-emitting unit in which heat is produced by air moving around a metal surface.

The numbers here

Modern cast iron boiler — projected life span

25 to 30 years

Expansion, and what happens when it is not handled

Water expands as it heats, and a sealed loop has nowhere to put the extra volume. The expansion tank holds a cushion of air that compresses to absorb it. Expansion tanks are only needed for hot water systems — a steam system has steam space of its own.

YOU HAVE MISSED THIS 2 TIMES

If the relief valve on a hot water heating system is discharging water, which of the following would most likely be the cause?

The system has a waterlogged or undersized expansion tank.

A waterlogged tank has lost its air cushion, so it is full of water and cannot compress. The next time the system heats up the pressure has nowhere to go and the relief valve does its job. The valve discharging is the symptom; the tank is the fault.

Automatic pressure relief valve	Should be located directly on the boiler, so nothing — no valve, no fitting — can come between the pressure and its escape.
Pressure regulator valve	A valve that limits the water pressure on certain heating equipment, on the supply side.

YOU HAVE MISSED THIS 2 TIMES

Which of the following statements about distribution piping is true?

In two-pipe distribution, the cold water leaving the radiator does not mix with the hot water.

One pipe feeds and returns through the same main, so each radiator gets water already cooled by the ones before it. Two-pipe keeps supply and return separate, so every radiator receives water at full temperature. That is the whole difference and the reason two-pipe is better.

Say it back

Cover the page. Answer each one out loud.

1. Why does a closed hot water system give even heat?
2. What is a hydronic system, and is steam one?
3. How do you tell a forced hot water system from a gravity one?
4. What is a tankless coil system, and what is a convector?
5. Cast iron boiler life span?
6. Why does a hot water system need an expansion tank, and does steam?
7. The relief valve is discharging. Most likely cause, and why?
8. Where must the pressure relief valve be located, and why there?
9. What is the advantage of two-pipe distribution?